

VIDEO COMPUTER SYSTEM™



OWNER'S MANUAL

MODEL CX2600 · FUJINET NETWORK CARTRIDGE



A FujiNet Project Publication

IN A HURRY?

Everything in this booklet is in the numbered sections that follow. If you have played with a FujiNet before, this is all there is to it:

- A. With the console **OFF**, push the FujiNet cartridge into the slot, label toward you. Plug a joystick into **LEFT CONTROLLER**.
- B. Slide the **POWER** switch **ON**. The FujiNet menu appears.
- C. Pick your WiFi network with the joystick and press the **red button** ("fire"). Spell the password on the on-screen keyboard. (Section 6.)
- D. On the **FN HOSTS** screen, move to **apps.irata.online** and press fire. Open **Atari_2600**.
- E. Move to a game and press fire. It loads and starts by itself.
- F. To come back to the FujiNet menu, press the **RESET button on the cartridge** (not the console's GAME RESET switch).

FujiNet Video Computer System Owner's Manual. First edition, 2026.

Typeset after the 1977 *Video Computer System Owner's Manual* as a tribute. FujiNet is an independent, open-source project and is not affiliated with, endorsed by, or sponsored by Atari. "Atari", "Video Computer System", "Combat", "Video Olympics" and "Dodge 'Em" are trademarks of their owners; "Dragster" and "Tennis" are Activision titles; "M.U.L.E." is a trademark of its owner. They are used here only to identify the console and the games this booklet is about.

The cartridge shell is a re-model of "Atari 2600 Cartridge Shell – Easy Print" by norm8332 (Thingiverse 1790785, CC BY). The pictures of the cartridge and the console are drawn from the FujiNet hardware design files; the screens are taken from the FujiNet cartridge running in MAME against a FujiNet adapter.

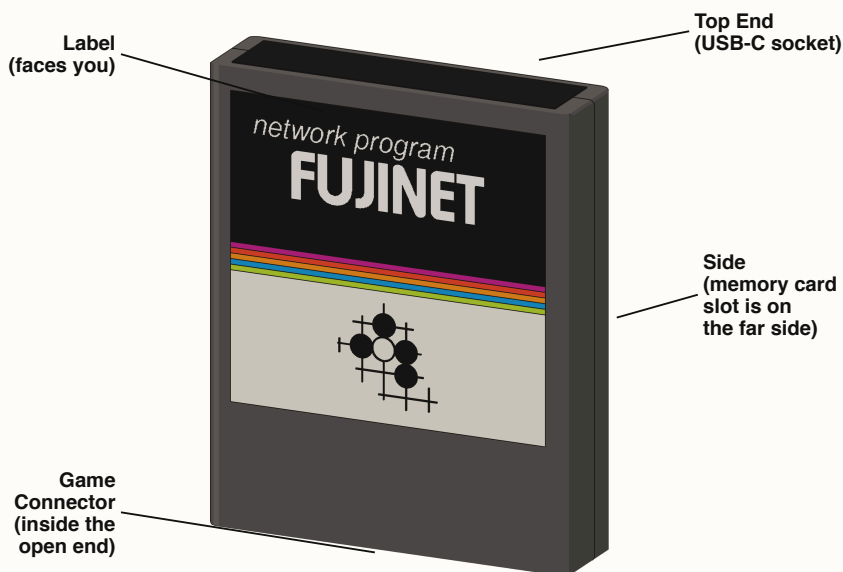
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The technical side of the cartridge – how it answers a console that has no network port, and how to write programs for it – is told in the *FujiNet Video Computer System Programmer's Handbook* and in *FujiNet for the Atari 2600: Theory of Operation*. This booklet tells you how to use it.

UNPACK CARTRIDGE



When you take your **FujiNet Network Cartridge** from its box, you should have:

- The FujiNet cartridge
- This Owner's Manual

The FujiNet cartridge goes in your console's Game Program slot like any other cartridge. Inside it is a complete network adapter: a WiFi radio, a memory-card reader and two small computers. It lets your Atari play games that are kept on the Internet, on a memory card, or on a computer in your home – and play some of them against people far away.

What You Also Need

- **An Atari 2600** of any model – the original six-switch console, the four-switch model, or the 2600 Jr. An Atari 7800, which plays 2600 cartridges, works too. FujiNet is made for **NTSC** (North American) consoles and televisions.
- **A joystick**, plugged into **LEFT CONTROLLER**. FujiNet's menus are worked with the joystick and its red button.
- **Your WiFi network's name and password**. FujiNet uses the 2.4 GHz band, which nearly every home router offers.

You may also want, though you do not need them to start:

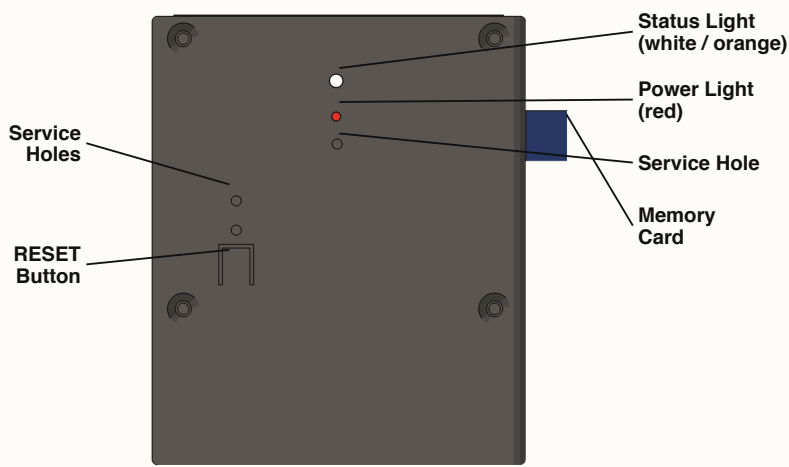
- **A microSD memory card**, formatted FAT32 (most cards come that way), to keep games on.
- **A USB-C phone charger and cable**, for extra power – see Section 4.

NOTE: FujiNet makes no changes to your console. Take the cartridge out and your Atari is exactly as it was.

Your FujiNet draws its menus in plain white letters on a black screen, and the games it loads draw whatever they like. As with any video game, turn your TV's brightness and contrast down a little when you play for a long time, and turn the console OFF when you are done.

KNOW YOUR CARTRIDGE 2

The cartridge has two faces. The **label** faces you, toward the front of the console. The **back** faces the back of the console, and carries everything you can press or see light up.



The back of the cartridge – the side that faces the back of the console.

The Lights

LIGHT	WHAT IT MEANS
White , steady	FujiNet is connected to your WiFi network.
Orange , flickering	FujiNet is busy: reading the network or the memory card, or sending a game to the console. It goes back to white when it is done.
Status light off	FujiNet is not connected to WiFi (yet).
Red	The cartridge has power and is running.

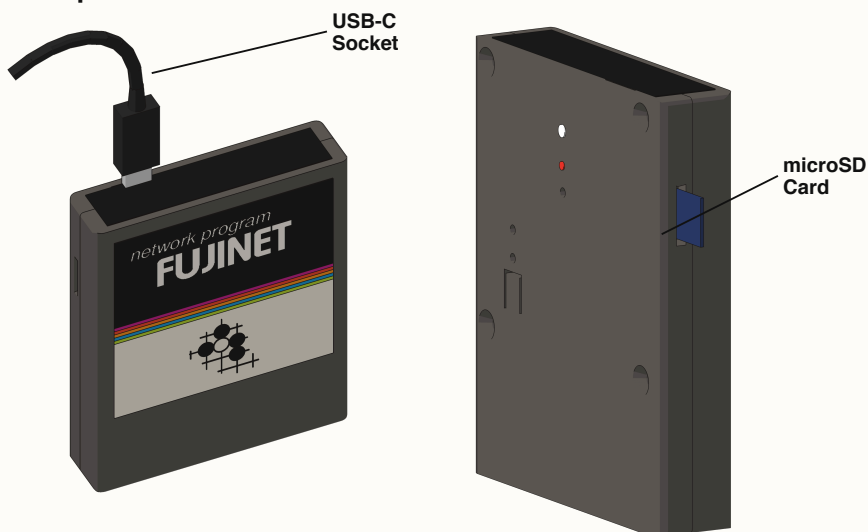
The lights face the back of the console. Lean over the top of the console to see them.

The RESET Button

The **RESET** button on the back of the cartridge restarts the cartridge and brings back the FujiNet menu, whatever game is running. It is the way home. It is not the same as the console's **GAME RESET** switch, which belongs to the game you are playing.

IMPORTANT: The three small service holes are for loading new firmware into the cartridge. Do not push anything into them in everyday use.

The Top End and the Side

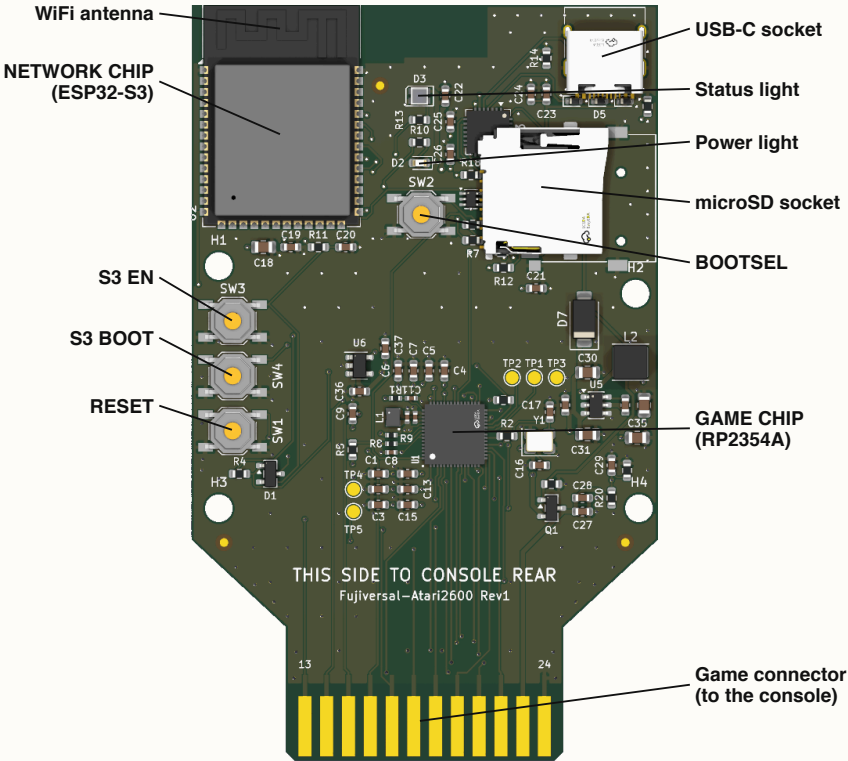


USB-C socket (top end). The cartridge normally runs on power from the console. A USB-C phone charger plugged in here powers it too. Power from the charger never flows back into the console, so it is safe to leave it plugged in. The same socket is used to load new firmware.

Memory card slot (side). A microSD card pushes into the slot in the side of the cartridge – the left end as you sit at the console – with its printed side toward the back of the cartridge, contacts first. Push it in until it clicks; push again to let it out. With the cartridge in the console you may find a fingernail or a guitar pick helps.

A Look Inside

You never need to open the cartridge, but if you are curious, this is what is inside. The board faces the back of the console, so this is the side the console's back sees.

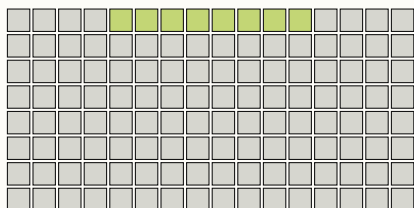


The Fujiversal-Atari2600 board, Rev1. S3 EN, S3 BOOT and BOOTSEL are the service buttons behind the three small holes.

3 HOW IT WORKS

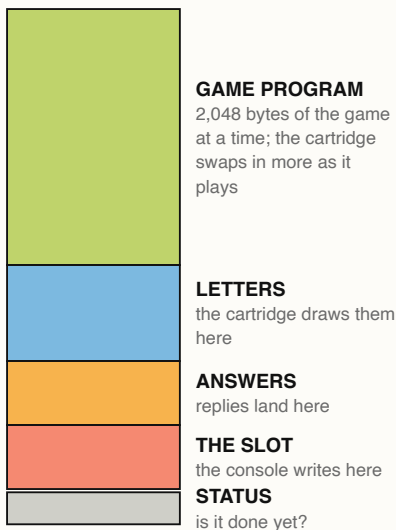
Your 2600 was built in 1977 to play cartridges. It has no network port, no keyboard, no letters of its own, and only **128 bytes** of memory – fewer than the letters in this paragraph. A FujiNet cartridge puts it on the Internet anyway. Here is the trick, in pictures.

THE CONSOLE'S MEMORY



All 128 bytes of it, one square to a byte. A FujiNet program borrows only a handful  to talk to the cartridge; everything else stays in the cartridge.

THE CARTRIDGE'S 4,096 BYTES



The console cannot hold a web page, a list of games or even a long name. So FujiNet keeps all of that in the **cartridge**, in the 4,096 bytes of address space every cartridge has, and lets the console look at it. Most of that space is the game itself. The rest is FujiNet's: a **slot** the console writes questions into, a **window** where the answers appear, and pages of **letters** the cartridge draws for it.

	THE CONSOLE, 1977	THE CARTRIDGE, 2026
Computers	one 6507, about 1.2 million steps a second	two (the game chip and the network chip), each a hundred times faster
Memory	128 bytes	over a million bytes, and 8 million more beside
Storage	the 4,096 bytes a cartridge shows it	18 million bytes of its own, and the memory card
Network	none	WiFi, and through it the whole Internet

How a Question Gets Answered

1 ASK

The game writes its question into **the slot**, one byte at a time. A cartridge is only meant to be read – the connector does not even have a wire that says “write”. But FujiNet’s game chip watches every address and every byte that crosses the connector, and it catches the question as it goes by.



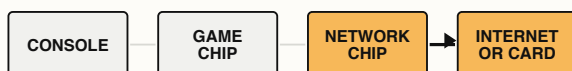
2 PASS IT ON

The game chip hands the question to the network chip beside it.



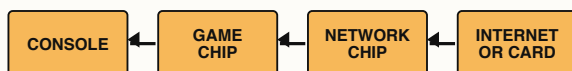
3 FETCH

The network chip does the work a 1977 console never could: it joins your WiFi, reaches the server, or reads the memory card.



4 ANSWER

The answer goes into the cartridge’s **window**. To the console it looks as though the answer had been printed in the cartridge all along. It simply reads it.



Letters, and Whole Games

The letters. The 2600 has no letters at all: everything on the screen is drawn a line at a time as the TV’s beam sweeps past. FujiNet’s cartridge works out the shape of every letter for it, so all the console does is copy them to the screen – which is why every FujiNet screen has the same square, friendly lettering, twelve letters across and twenty-one lines down.

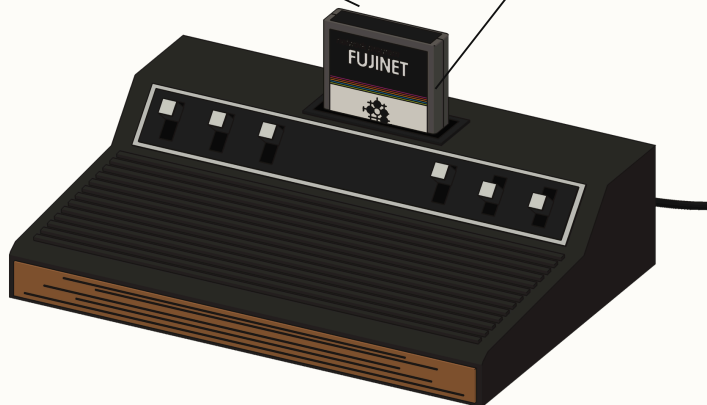
Whole games. When you load a game, FujiNet sends all of it, up to 32,768 bytes, into the cartridge’s own memory. Then the menu steps aside and the cartridge becomes that game’s cartridge, page-switching and all. The console never knows the difference.

The full story – every address, every command, and how to write your own FujiNet programs for the 2600 – is in the *FujiNet Video Computer System Programmer’s Handbook*, sections 3 to 6, and in *FujiNet for the Atari 2600: Theory of Operation*, Parts I, III and V.

4 INSERT CARTRIDGE

FujiNet Cartridge

Game Program Slot



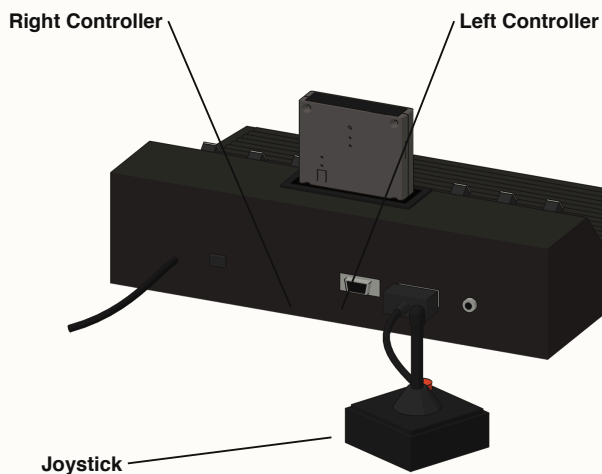
- Check to see that the **POWER** switch on the console is **OFF**.

IMPORTANT: To protect your console and the cartridge, the console should be **OFF** whenever you put a cartridge in or take one out.

- Hold the FujiNet cartridge so its label faces you and reads right side up.
- Push it **carefully** into the slot in the middle of the console until it is firmly seated. Do **not** try to force it.

When you take the cartridge out, pull it straight up, and check first that the **POWER** switch is **OFF**.

Plug in a Joystick



The back of the console. The joystick goes in LEFT CONTROLLER.

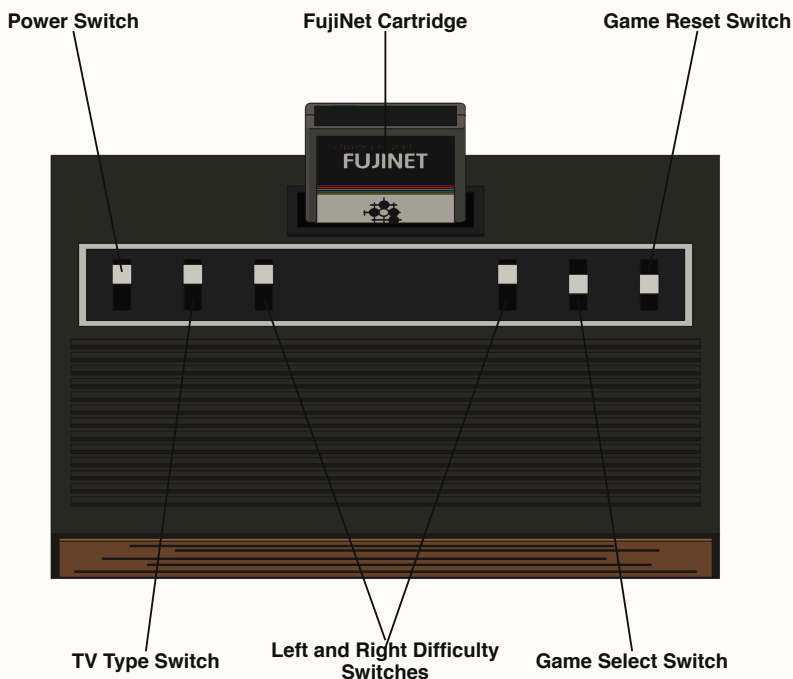
Plug a joystick into the **LEFT CONTROLLER** jack on the back of the console. The FujiNet menu and every FujiNet game listen to the left joystick. The plug fits only one way; do not force it.

NOTE: Some games want a second joystick, or paddles. The game will tell you; FujiNet does not mind what is plugged in.

Extra Power

The cartridge runs on power from the console. If your console is a 2600 Jr or a 7800, or if the picture jumps or the FujiNet menu starts over by itself when the WiFi connects, plug a **USB-C phone charger** into the top of the cartridge. The cartridge takes what it needs from the charger and never sends any of it back to the console.

5 TO START PLAY



- Turn on your TV, set its volume low, and tune it to **channel 3**, just as for any Atari game. (If your console is connected some other way, use the input it always uses.)
- If your TV is black and white, slide the **TV TYPE** switch to **B·W**. Otherwise leave it at **COLOR**.
- Slide the **POWER** switch **ON**. The FujiNet menu appears.

The first time you turn it on, FujiNet looks for WiFi networks and asks you to choose yours. That is the next section. After that, it remembers, and goes straight to the list of places where games are kept (Section 7).

The Joystick in the FujiNet Menu

FujiNet's menus are twelve letters wide and need only the joystick and two switches. They all work the same way:

IN EVERY FUJINET MENU

>	The arrow at the left marks the line you are on.
▲▼	Move up and down the list.
◀▶	Turn the page, when a list is longer than the screen.
FIRE	Choose the line you are on.
GAME SELECT	Open the ACTIONS menu for this screen: a short list of the things you can do here. GAME SELECT again closes it without choosing.

The menu does not use the GAME RESET switch, the difficulty switches or the TV TYPE switch. The game you load decides what those do.

NOTE: The RESET button on the cartridge is not the console's GAME RESET switch. GAME RESET belongs to the game. The cartridge's RESET always brings you back to the FujiNet menu.

Hold the Joystick Right Way Up

The red button goes at the top left, away from you, with the cord leading toward the TV – the way it sits naturally in your hand.

JOIN YOUR NETWORK

6

The first time you turn on the console with the FujiNet cartridge in it, FujiNet looks for WiFi networks nearby. The screen may go dark for a few seconds while it listens. Then it lists what it heard.



PICK NET

- A. Push the joystick up or down until the > is beside the name of your network.
- B. Press the red button. The keyboard appears, titled **PASSWORD**.
- C. Spell your password (see *The Keyboard*, below).
- D. Press **GAME SELECT**, move to **ACCEPT** and press the red button.

Names longer than eleven letters are cut short. If your network is not in the list – some routers hide their names – choose **OTHER...** at the bottom. FujiNet asks for the network's name (**ENTER SSID**) first, and then for the password.



Joining



It did not work

FujiNet now tries to join. The screen says **CONNECTING** and **PLEASE WAIT**; this can take up to about forty seconds. When it is done, the status light on the cartridge turns white and the list of hosts appears (Section 7). FujiNet remembers your network, so next time it joins by itself.

If the screen says **WIFI FAILED**, the password was not right or the router is too far away. Press **GAME SELECT** to go on without WiFi, then choose **WIFI** from the ACTIONS menu on the hosts screen to try again.

NOTE: You can skip WiFi altogether: press **GAME SELECT** at **PICK NET**. FujiNet still plays games from its memory card.

The Keyboard

FujiNet has no keyboard to plug in, so it draws one. Move the ^ pointer under a character with the joystick and press the red button to type it. The top line shows what you have typed so far; **LEN** counts the letters.



The keyboard

The bottom-right key, <, erases the last letter.

GAME SELECT opens the keyboard's ACTIONS:

- **ACCEPT** – done; use what is typed.
- **CANCEL** – leave it as it was before.
- **CLEAR** – start over.

A password can be up to 63 letters, a network name up to 32.

IMPORTANT: Passwords care about capitals and small letters. FujiNet's screen draws them all the same way, so use this chart: the top letters on the keyboard are **CAPITALS** and the bottom three rows are small letters.

SPACE	!	"	#	\$	%	&	'	(	)	*	+
,	-	.	/	0	1	2	3	4	5	6	7
8	9	:	;	<	=	>	?	@	A	B	C
D	E	F	G	H	I	J	K	L	M	N	O
P	Q	R	S	T	U	V	W	X	Y	Z	[
\	]	^	_	`	a	b	c	d	e	f	g
h	i	j	k	l	m	n	o	p	q	r	s
t	u	v	w	x	y	z	{		}	~	ERASE

What each key types. The shaded rows are small (lower-case) letters, even though the screen shows them as capitals.

HOSTS AND THE BROWSER

7

A **host** is a place where games are kept: the memory card in the cartridge, a server on the Internet, or a computer in your home. FujiNet remembers eight of them, and lists them on the **FN HOSTS** screen every time you turn on.



A new FujiNet comes with three hosts already filled in:

- **SD** – the microSD card in the side of the cartridge.
- **tnfs.fujinet.online** – the FujiNet project's own library.
- **apps.irata.online** – the library that keeps the 2600's network games.

The other five say **(EMPTY)**. Names are cut to eleven letters, so *apps.irata.online* shows as **APPS.IRATA**.

Move to a host and press the red button to look inside it.

FN HOSTS, as it comes



What your TV shows. FujiNet's twelve-letter menus sit in a narrow column in the middle of the screen. The screens in this booklet show that column close up. (This one has a few more hosts filled in.)

Adding a Host

- A. Move the > to an (EMPTY) line.
- B. Press **GAME SELECT**. Choose **RENAME**.
- C. The keyboard appears, titled **HOST NAME**. Spell the server's name or address, for example 192.168.1.20 Or games.local.
- D. Press **GAME SELECT** and choose **ACCEPT**.

To change a host, RENAME it the same way. A host name can be up to 31 letters. Most hosts are **TNFS** servers, and a plain name means TNFS. FujiNet can also reach other kinds of server if you spell the kind in front:

TYPE THIS	FOR
SD	the memory card in the cartridge
games.local OR 192.168.1.20	a TNFS server – the usual kind
smb://server/share	a Windows or Samba file share
ftp://server	an FTP server
http://server/path	a web server (https:// too)



The ACTIONS menu

The Hosts Menu

GAME SELECT on the FN HOSTS screen opens its ACTIONS:

- **INFO** – your network and FujiNet's version (Section 10).
- **RENAME** – fill in or change the host you are on.
- **WIFI** – choose a WiFi network again (Section 6).
- **LOBBY** – the FujiNet Game Lobby (Section 11).

Up and down choose; the red button does it; **GAME SELECT** closes the menu and does nothing.

The Browser

When you open a host, **FN BROWSE** shows what is in it, fourteen names to a page. The second line shows where you are, ending in the folder's own name.

Folders end in  and come first; games follow, A to Z.



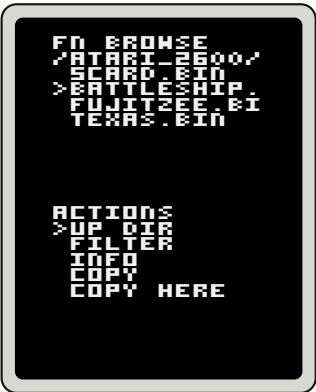
The top of *apps.irata.online*



Inside *Atari_2600*

IN THE BROWSER

	Move up and down the page.
	Next page (when there is more).
	Previous page. On the first page, go back up one folder; at the top of the host, go back to FN HOSTS.
	On a folder, open it. On a game, load it (Section 8).
	Open the browser's ACTIONS menu.



The browser's ACTIONS

- **UP DIR** – back up one folder.
- **FILTER** – show only the names that match a pattern. Spell it on the keyboard: * stands for any letters and ? for any one letter, so B* shows only games beginning with B, and *.BIN only .BIN files. Folders are always shown. An empty filter shows everything.
- **INFO** – your network and FujiNet's version.
- **COPY, COPY HERE** – copy a game to another host (Section 9).

Long names are cut to eleven letters; the full name is still what FujiNet loads. The small .CFG files some games need (Section 8) are kept out of the list.

8 LOADING GAMES

To load a game, find it in the browser and press the red button. FujiNet sends the whole game into the cartridge and starts it – there is nothing else to do.



Loading



It would not load

BOOTING shows the end of the game's name and a bar that fills as it loads. **DO NOT POWER** off while it does. Then the screen changes, and the game starts just as if its own cartridge were in the slot.

If it says **BOOT FAILED**, the game is one FujiNet cannot play (see the table on the next page), or the file is damaged. After a few seconds you are back in the browser.

IMPORTANT: To leave a game and go back to the FujiNet menu, press the **RESET** button on the cartridge. Turning the console off and on again does the same. The console's **GAME RESET** switch only restarts the game.

From the Internet

The FujiNet project keeps 2600 games on the Internet, ready to load. Nothing needs to be copied:

HOST	FOLDER	WHAT IS THERE
apps.irata.online	Atari_2600	The FujiNet network games: 5 Card Stud, Texas Hold'em, Battleship and Fujitzee (Sections 12–15)
apps.irata.online	mule / atari-2600	M.U.L.E. (Section 16)
tnfs.fujinet.online		The FujiNet project's library for every computer and console

Open the host, open the folder, and choose the game.

From the Memory Card

- On your computer, copy your games onto a microSD card formatted FAT32. You may put them in folders.
- With the console OFF, push the card into the side of the cartridge until it clicks (Section 2).
- Turn on. Open the **SD** host, and choose a game.

From Your Own File Server

A computer in your home can serve games to FujiNet with a small free program called a **TNFS server**. There are versions for Windows, Mac, Linux and the Raspberry Pi; see fujinet.online for the current one.

- Install the TNFS server, and point it at a folder of games.
- Find your computer's network address – something like 192.168.1.20.
- On FN HOSTS, RENAME an (EMPTY) host to that address (Section 7).
- Open it. Your folder appears in the browser.

A Windows share (smb://) or an FTP or web server works the same way.

Which Games Will Load

FujiNet loads a game into its own memory and then plays the part of that game's cartridge, page-switching included. Most of the 2600 library loads just by choosing it:

GAME SIZE AND KIND	FOR EXAMPLE
2K and 4K games	Combat, Adventure, Pitfall!, Space Invaders
8K (F8)	Asteroids, Ms. Pac-Man, Raiders of the Lost Ark
12K (FA, CBS RAM Plus)	Omega Race, Mountain King, Tunnel Runner
16K (F6)	Solaris, Midnight Magic
32K (F4)	Fatal Run
Games with an extra Super Chip (F8SC, F6SC, F4SC)	Dig Dug, Crystal Castles, Off the Wall – found by themselves

A few kinds of game cannot be told apart by their size. They load when a small text file sits beside them, with the **same name** ending in .CFG, holding one word: the kind.

WORD	KIND	FOR EXAMPLE
E0	Parker Brothers 8K	Frogger II, Montezuma's Revenge, Popeye, Gyruss
FE	Activision 8K	Decathlon, Robot Tank
UA	UA Limited 8K	Pleiades, Funky Fish
CV	CommaVid	Magicard, Video Life

For example, MONTEZUMA.BIN and a file MONTEZUMA.CFG containing just E0. FujiNet hides .CFG files in the browser and copies them along with their games.

File names. FujiNet loads files whose names end in .BIN or .ROM. Many 2600 game files end in .A26; rename them to .BIN. Games may be up to 32K.

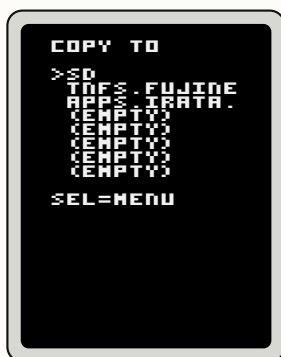
Not yet supported: Pitfall II and other games with an extra chip of their own (DPC), Starpath Supercharger tapes, Tigervision and other games larger than 32K or with their own page-switching (3E, 3F, F0, EF, DF, BF, SB), and the modern ARM-powered homebrews.

NOTE: FujiNet is made for NTSC consoles and TVs. PAL games may roll or lose their colour.

9 COPYING GAMES

FujiNet can copy a game from one host to another – from the Internet to your memory card, say, so that you can play it without a network.

- In the browser, move the **>** to the game you want to copy. Press **GAME SELECT** and choose **COPY**.
- The hosts screen comes back, titled **COPY TO**. Choose the host to copy to – **SD** for the memory card – and press the red button.
- Find the folder the game should go in. Press **GAME SELECT** and choose **COPY HERE**.



Step B



Copying



Done

COPYING shows where the game is coming from and where it is going. When it says **COPIED**, press the red button (or wait a moment) to go back to the browser. If it says **COPY FAILED**, the memory card may be full, or missing, or the host may not let you write to it – the Internet libraries, for one, are read-only. A game's .CFG file, if it has one, is copied along with it.

INFO AND THE WEB CONTROL PANEL

10



FN INFO

Choose **INFO** from the ACTIONS menu on the hosts screen or in the browser to see how FujiNet is connected:

LINE	WHAT IT IS
SSID	the WiFi network FujiNet joined
IP	FujiNet's address on your network
GATEWAY	your router's address
DNS	the name server it uses
MAC	FujiNet's own hardware number (only the start fits)
VERSION	the FujiNet firmware version

Press the red button to go back.

The Web Control Panel

Everything FujiNet knows can also be seen and changed from a computer, tablet or phone on the same network. Type FujiNet's **IP** address from the INFO screen into a web browser – for example <http://192.168.1.73/>. On many networks <http://fujinet.local/> works too.

The control panel lets you change the WiFi network, fill in or change the eight host slots with a real keyboard, and see the adapter's settings and version. It is the easiest way to type a long host name.

Setting WiFi from the Memory Card

If you would rather not spell a password with the joystick, FujiNet can read its settings from the memory card. Make a plain text file called `fnconfig.ini` in the top folder of the card, holding:

```
[WiFi]
enabled=1
SSID=YourNetworkName
passphrase=YourPassword
```

Put the card in the cartridge and turn on. FujiNet reads the file each time it starts. (Everything you change in the menu or the control panel is written back to the same file, so the card keeps your hosts as well.)

THE GAME LOBBY

11

The **FujiNet Game Lobby** is a list of every network game that has a table open right now, on every kind of computer and console FujiNet works with. Pick a table and the Lobby loads the right game and sends you straight to it.

To open the Lobby, press **GAME SELECT** on the FN HOSTS screen and choose **LOBBY**. FujiNet loads it from the Internet.



The top line says **LOBBY** and which page you are on. Each game has a coloured band; under it are its tables, with how many players are sitting and how many seats there are – **2/8** is two players at a table for eight.

IN THE LOBBY	
	Choose a table.
	Previous or next page.
	Join that table: load its game and sit down.
	Change your name.
	Fetch the list again.

The bottom of the screen shows **SEL=** and your name – **GAME SELECT** is the switch that changes it. **RST=REFRESH** reminds you that **GAME RESET** asks for a fresh list.

Your Name

FujiNet keeps one player name for all of its games, so you type it once. The Lobby asks for it the first time (**ENTER NAME**), and every FujiNet game uses the same one. On the name keyboard the joystick moves, the red button types the letter you are on, **GAME SELECT** erases, and **GAME RESET** is done. Names are up to eight letters and numbers.

What You May See

SCREEN SAYS	WHAT IS HAPPENING
LOADING	Fetching the list of tables.
NO SERVERS	No game has a table open just now. Try again later, or load a game yourself from the browser – its own table list works without the Lobby.
JOINING, MOUNTING	Getting your game ready.
DO NOT STOP	Loading the game. Leave the console on.
ERR and a number	Something on the network did not answer. Press GAME RESET to try again.

NOTE: Every FujiNet game in this booklet can also be loaded straight from the browser (Section 8). Each shows its own list of tables, so you can play with or without the Lobby.

Up to eight players around one table, on any computer or console FujiNet plays on – and if there are not enough people, the house's robots fill the seats. Each hand you get one card face down and four face up, and the best poker hand takes the pot.

Loading the Game

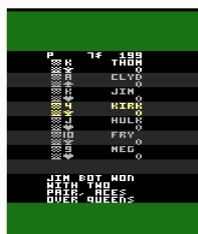
Choose a 5 Card Stud table in the Lobby (Section 11), or load it from the browser: host **apps.irata.online**, folder **Atari_2600**, game **5CARD.BIN**.

The first time, the game asks **YOUR NAME**. Move around the letters with the joystick and press the red button to type; **OK** finishes, **DEL** erases, **SPC** is a space. (If you have already given FujiNet a name, it just uses that.)

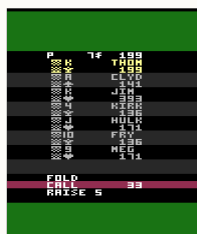
Then comes the list of tables. **The Basement** and **The Den** are for people; the **AI Rooms** seat you with two, four or six robot players, which is the best way to learn. Move to a table and press the red button – **FIRE TO SIT**.



The tables



The hand is over



GAME SELECT held



TABLE MENU



The Table

- **Top line:** P and the pot, then \$ and your purse.
- **Eight seats**, two lines each: each card's rank over its suit, then the player's name and what they have bet. A face-down card is drawn as a hatched back until the showdown.
- A seat's **colour** tells you its state: whose turn it is, who has folded, who has left, and which seat is yours.
- **Bottom lines:** your moves when it is your turn. The bar marks the move you are on, and the number on it is the seconds you have left. Otherwise **WAITING ON** and whose turn it is.

5 CARD STUD

	Choose a move (or a table, or a letter).
FIRE	Make that move. In the table list, sit down.
GAME SELECT held	Every seat shows its purse instead of its bet.
	Ask the server now, instead of waiting.
 or GAME RESET	The TABLE MENU: RESUME, HOW TO PLAY, or LEAVE TABLE.

Your First Hand

- Everyone puts **1** in the pot to start (the ante). Each player is dealt one card face down and one face up.
- The player showing the lowest card must **POST 2** to begin the betting. Going round the table, each player may **CALL** (match the bet), **BET** or **RAISE** (make it bigger), or **FOLD** (drop out of this hand).
- A third, fourth and fifth card are dealt face up, each followed by another round of betting. Bets are **5** in the early rounds and **10** later; a round allows no more than three raises.
- When the betting is done, the hidden cards are turned over. The best poker hand wins the pot, and the end-of-hand message runs along the bottom.
- About twelve seconds later the next hand is dealt.

Everyone starts with **\$200**. You have about **forty seconds** for each move. If the clock runs out, the move under the bar is sent for you – and the bar never starts on FOLD, so you will not drop out by accident.

NOTE: To leave, use **LEAVE TABLE** from the **TABLE MENU**. It gives your seat back properly, so the others are not left waiting for you.

13 TEXAS HOLD'EM

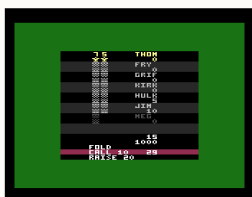
The poker game everyone knows, for up to eight players with robots to fill empty seats. Each player gets two cards of their own, five cards in the middle are shared by all, and the best five-card hand wins.

Loading the Game

Choose a Texas Hold'em table in the Lobby, or load it from the browser: host **apps.irata.online**, folder **Atari_2600**, game **TEXAS.BIN**. The name screen and the table list work just as in 5 Card Stud. Pick an **AI ROOM** to learn against robots, or **THE DEN** or **THE BASEMENT** to play people.



The tables



Your turn



The hand is over



TABLE MENU

The Table

Your seat is at the top. Under each player's name are their two cards (yours face up, the others' face down until the showdown) and their chips. Below the eight seats are the **shared cards** – three, then four, then five as the hand goes on – with the pot and your purse beside them. At the bottom are your moves, with the bar on the one you are about to make and the seconds you have left.

The controls are the same as 5 Card Stud's: up and down choose a move, the red button makes it, holding GAME SELECT shows everyone's purse, right asks the server now, and left or GAME RESET opens the **TABLE MENU**.

Your First Hand

- Two players put in the **blinds**, **5** and **10**, to start the pot. Everyone is dealt two cards face down.
- Going round the table, each player may **CALL**, **RAISE** or **FOLD** – or **CHECK**, if there is nothing to call.
- Three shared cards are dealt (the **flop**), then another betting round; then a fourth (the **turn**) and a fifth (the **river**), each with its own betting round.
- At the showdown, each player makes the best five-card hand from their two cards and the five shared ones. The best hand takes the pot.

Everyone starts with **\$1,000**. As in 5 Card Stud you have about forty seconds a move, and the next hand follows by itself.

14 BATTLESHIP

Two to four fleets on one ocean. Hide your five ships, then take turns firing – and every shot you fire lands on **every** enemy board at once. The last fleet afloat wins.

Loading the Game

Choose a Battleship table in the Lobby, or load it from the browser: host **apps.irata.online**, folder **Atari_2600**, game **BATTLESHIP.BIN**. Give your name if it asks (**YOUR NAME**: the red button types, **OK** finishes). The table list shows each table and how many **SEATS** are taken; **FIRE JOINS**. The **AI** tables give you one, two or three robot admirals; **Cape Fuji** and **High Seas** are for people.



The tables



Waiting to start



Two fleets

In the waiting room, press the red button when you are **ready**. The game starts when everyone is ready, or about thirty seconds after the first player sits down.

Hiding Your Ships

You have five ships, five, four, three, three and two squares long. The game starts you with a fleet already laid out, so you can simply press the red button five times to keep it. To move a ship instead:



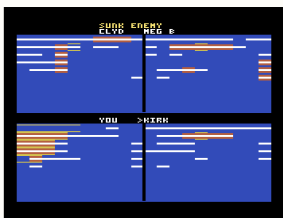
PLACE SHIP and the number of the ship you are placing are at the top; your ships are outlined in gold on your own board. The hints at the bottom say **SEL TURNS** and **FIRE PLACES**.

PLACING SHIPS – PLACE SHIP

Joystick	Move the ship that is being placed.
GAME SELECT	Turn it (SEL TURNS).
FIRE	Leave it there and go on to the next ship (FIRE PLACES).

A ship that would hang off the edge says **WON'T FIT**; one that would sit on another says **SHIPS TOUCH**. Move it and try again.

Battle



Every board is on the screen. With two players there are two big boards; with three or four, four boards – yours at the bottom left, the others going round clockwise. **Gold** marks your ships and your aim, a **red** square with a white middle is a hit, and a **white** dash is a miss. Under the boards, each fleet's ships show as **#** while afloat and **=** when sunk.

The top line tells you what happened last, whose turn it is, and the seconds you have left – **MISS YOU 45, HIT ENEMY, SUNK YOU 12** – and finally **YOU WIN!** or the winner's name.

IN BATTLE

Joystick	Aim.
FIRE	Fire at that square, on every enemy board at once.
GAME SELECT	Ask the server now, instead of waiting.
GAME RESET	The GAME MENU: RESUME, HOW TO PLAY, or LEAVE TABLE.

You have **45 seconds** a turn. A square you have already fired at on every board is refused with a buzz, so you cannot waste a shot.

FUJITZEE 15

The dice game. Roll five dice up to three times a turn, keep the ones you like, and score the result in one of thirteen boxes on your card. After thirteen rounds the highest total wins. Up to six players, with robots on hand.

Loading the Game

Choose a Fujitzee table in the Lobby, or load it from the browser: host **apps.irata.online**, folder **Atari_2600**, game **FUJITZEE.BIN**. Give your name if it asks (**YOUR NAME; RST=DONE** – here GAME RESET finishes). On **PICK A TABLE**, choose a table and press the red button (**FIRE=JOIN**). The **Bar** and **Kitchen Table** are for people; the **AI Rooms** add two or four robots. In the waiting room, press the red button when you are ready.



Your Turn

The card fills the screen, one box to a line. The five dice sit near the bottom, with **ROLL** and the rolls you have left beneath them.

- The pointer starts on **ROLL**. Press the red button to roll.
- Move left and right along the dice. The red button **holds** a die (drawn as brackets, with a mark beneath); press again to let it go.
- Roll again – up to three rolls in all.
- Push up onto the card. **Green** lines show what this roll would score in each box. Move to the box you want and press the red button.

Pressing the red button over and over plays a legal, if unwise, game: the pointer starts on ROLL, and after the last roll it waits on the box that pays best.

FUJITZEE

Joystick	Left and right along the dice; up onto the card; up and down the boxes.
FIRE	Roll, hold a die, or score a box. In the waiting room, ready .
GAME SELECT	The MENU: RESUME, HOW TO PLAY, LEAVE TABLE .
GAME RESET	Ask the server now.
LEFT DIFFICULTY at A	Show everyone's standings, for as long as it stays at A.

The Card

BOX	SCORES
ONE through SIX	The total of the dice showing that number
UP	Your top-half total, and a bonus of 35 once it reaches 63
SET 3, SET 4	Three or four of a kind: the total of all five dice
FULL HSE	Three of one number and two of another: 25
SM STRT	Four in a row: 30
LG STRT	Five in a row: 40
FUJITZEE	All five the same: 50
CHANCE	Anything: the total of all five dice

Lines in **blue** are boxes already filled; **yellow** is your pointer (or the box a robot has just taken). You have **45 seconds** a turn. At **GAME OVER** the standings show who won.



Waiting for everyone

Game over

M.U.L.E. 16

The classic game of settling the planet Irata, for four colonists on any FujiNet machine at all – the other three may be on an Atari computer, an Apple, a PC, an Intellivision or another 2600, and the computer takes any empty seat. Twelve months to claim land, outfit your M.U.L.E.s, and trade your way to the richest colony.



The colony room



The town



The map

Loading the Game

Choose **M.U.L.E.** and the room **Planet IRATA** in the Lobby, or load it from the browser: host **apps.irata.online**, folder **mule**, then **atari-2600**, game **MULE2600.BIN**. On the title screen press the red button (**FIRE: START**), give your name if it asks, and pick a table (**FIRE: JOIN**; **RIGHT: RELOAD** fetches the list again). In the colony room, choose your race with the joystick and press the red button when you are ready.

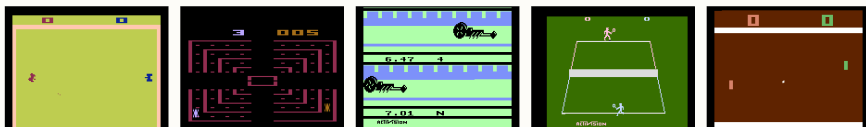
M.U.L.E.	
Joystick	Walk; choose; move your auction line; declare (up to sell, down to buy).
FIRE	Start, join and ready; claim a plot; begin your turn; install a M.U.L.E., take a sample, catch the Wampus; collude; sit out a declaration.
GAME SELECT	End your turn.
GAME RESET held for a second	Leave the table. The computer takes over your colonist.
LEFT DIFFICULTY at A	Sound off.

The 2600 draws the map and the town with letters. On the map, **F**, **E**, **S** and **C** are Food, Energy, Smithore and Crystite M.U.L.E.s, and each owner's colour

frames their land. In the town, **C S E F** are the outfitters and **A L P** the assay office, the land office and the pub. Messages run along the bottom line.

The whole game – the colonists, the land, every phase of a month, the auctions, every fortune and catastrophe – is in the *M.U.L.E. Player's Guide*, Atari 2600 edition, from the FujiNet project.

Some of the best games ever made for the 2600 were for two players sitting side by side. The FujiNet project is teaching five of them to reach across the Internet instead – the same games, unchanged in how they look and play, with your opponent at their own console, in their own home.



Combat · Dodge 'Em · Dragster · Tennis · Video Olympics

NOTE: These five are being finished now. As each one is ready it will appear in the FujiNet Game Lobby (Section 11). What follows is how they will play.

How a Match Works



Both consoles run the whole game. All that travels between them is what each player does with the controller – fifteen times a second, through a **relay** on the Internet that pairs the first two players who arrive. Each console waits the same short moment (about an eighth of a second) before acting on **both** players' moves, so neither of you is ever ahead. If the other player's moves are late, the picture holds still for a moment rather than going wrong.

- Each player uses **their own** joystick in **their own** console's **LEFT CONTROLLER** jack.
- **GAME SELECT** and **GAME RESET** work from either console, and both games obey them at the same instant.
- The **TV TYPE** switch stays your own: you can play in colour while your opponent plays in black and white.
- If the two games ever disagree, they press GAME RESET together, by themselves, and start the round again.

What You Will See



Finding the relay

Waiting for a player

Paired: your opponent's name

No network

CONNECTING, then **WAITING FOR AN OPPONENT** until someone joins, then **PLAYING** and your opponent's name – and the game begins. With no network, or no relay, the screen says **NO NETWORK / LOCAL PLAY** for a moment, and then the old game starts just as it always did, for two players on one console: press **GAME RESET** to play. If your opponent leaves, the screen says **OPPONENT HAS LEFT** – press **GAME RESET** to look for another.

The Games

COMBAT (ATARI, 1977)

The first player to arrive drives the **left** tank (or plane), the second the right. All 27 games are there: tanks, Tank-Pong, invisible tanks, biplanes and jets, guided or straight shots, open field, mazes and clouds. **Choose the game with GAME SELECT first, then press GAME RESET** – GAME SELECT is ignored once a battle has started. Each player's **own left difficulty switch** sets their own tank's handicap. A game lasts about two minutes and sixteen seconds.

DODGE 'EM (ATARI, 1980)

Over the network it is **game 3**, the two-player game: one of you drives the car collecting the dots while the other drives the crash car after them, and you swap every round, so both of you are busy all the time. The difficulty switches cross the wire too.

DRAGSTER (ACTIVISION, 1980)

The first player races in the **top** lane, the second in the **low** lane; the screen tells you which, and **JOY RIGHT TO STAGE**. Push the joystick right to stage. Pull it **left** to arm a gear change and let go to take the gear – letting go is also your launch, the instant the tree turns green. The red button is the throttle; hold it past the red line and the engine blows. GAME SELECT chooses the straight race or the race with a drift to steer against. The starting lights are shown a moment early so your launch is not late; gear changes still arrive a little after you make them, so network times run a touch slower than on one console.

TENNIS (ACTIVISION, 1981)

The screen tells you whether you are **PLAYER ONE** or **PLAYER TWO**. You swap ends every game, just as in real tennis. The red button serves; swings are automatic. GAME SELECT offers the two two-player games (the second is the slower one). Your own difficulty switch limits your racket's angles, and it crosses the wire.

VIDEO OLYMPICS (ATARI, 1977)

A **paddle** game: each player plugs paddles into their own console's **LEFT CONTROLLER** jack and uses the first paddle. The first player to arrive is the left side, the second the right. GAME SELECT steps through the 22 of its 50 versions that two players can play; your own difficulty switch crosses the wire.

How a 1977 game learns to play over the Internet without changing how it plays is a story in itself, told in the *PORTING* notes that come with each game's source code.

MAINTENANCE

18

Your FujiNet cartridge should bring many years of enjoyment. To keep it in good working condition, please remember the following points:

- Always turn the **POWER** switch on the console **OFF** before putting the cartridge in or taking it out.

- Don't try to force the cartridge into the console.
- Don't push anything into the three small service holes on the back of the cartridge.
- Push the memory card straight in and let it spring straight out. Don't pry at it.
- Use only a USB-C phone charger or a computer's USB port in the USB-C socket.
- Don't spill liquids on the cartridge or pour them into the console's slot.
- Don't drop the cartridge or leave it in great heat.
- Clean the outside of the cartridge with a soft, slightly damp cloth.

Keeping FujiNet Up to Date

FujiNet's firmware – the programs inside the cartridge – is improved all the time. New versions, and how to load them, are announced at **fujinet.online**. Loading new firmware does not change your hosts or your WiFi settings.

TROUBLE SHOOTING CHECKLIST

19

SYMPTOM	PROBABLE CAUSE AND REMEDY
No picture, or a grey or snowy screen.	• Console POWER switch not ON. • TV not on channel 3, or the TV/Game switch set to TV. • Cartridge not firmly seated. Turn the console OFF, push the cartridge in firmly, and turn it ON. • Cartridge put in while the console was ON. Slide POWER OFF and back ON.
The picture rolls, or jumps when WiFi connects; the FujiNet menu keeps starting over.	• The console cannot give the cartridge enough power. Plug a USB-C phone charger into the top of the cartridge (Section 4). • A PAL console or TV. FujiNet is made for NTSC.
The joystick does nothing.	• Joystick plugged into RIGHT CONTROLLER. Move it to LEFT CONTROLLER. • Plug not pushed fully home.
My network is not in the PICK NET list.	• Router too far away. Move the console nearer, or the router. • Network uses only 5 GHz. FujiNet needs 2.4 GHz. • Hidden network. Choose OTHER... and spell its name.
WIFI FAILED.	• Wrong password. Remember the bottom three rows of the keyboard are small letters (Section 6). Press GAME SELECT, then choose WIFI on the hosts screen and try again. • Or set WiFi from the memory card (Section 10).
Status light not white.	• FujiNet is not on your WiFi. See WIFI FAILED, above.
A host shows nothing, or an error such as E03 FF.	• Host name misspelled. RENAME it (Section 7). • The server is not running, or your Internet connection is down. • SD: no memory card, or it is not formatted FAT32.
BOOT FAILED.	• A game FujiNet cannot play yet (Section 8). • An 8K game that needs a .CFG file beside it (Section 8). • A damaged file. Copy it again.
The game starts, but is garbled or rolls.	• A PAL game. Use the NTSC version. • The wrong .CFG word for the game.
A game file is not in the browser.	• Its name does not end in .BIN or .ROM. Rename .A26 files to .BIN. • A FILTER is set. Choose FILTER and ACCEPT an empty one.

SYMPTOM	PROBABLE CAUSE AND REMEDY
I cannot get back to the FujiNet menu.	• Press the RESET button on the back of the cartridge, not GAME RESET. • Or turn the console OFF and ON again.
The Lobby says NO SERVERS.	• No tables are open just now. Load the game from the browser instead; it has its own table list.
A network classic stays at WAITING FOR AN OPPONENT.	• Nobody else has joined yet. Leave it waiting, or arrange a time with a friend.

As a rule of thumb, if something does not work, turn the console OFF, wait ten seconds, and turn it ON again.

Getting Help

FujiNet is made by a worldwide community of volunteers, and they are glad to help:

- fujinet.online – news, downloads and the FujiNet wiki
- github.com/FujiNetWIFI – the source for everything, and the place to report a problem
- discord.gg/7MfFTvD – the FujiNet Discord, where you can ask anything

20 PARTS LIST

The FujiNet cartridge is open hardware: the board, the shell and every program in it may be built, changed and shared. Its design files are kept in the **fujinet-hardware** collection on GitHub, under **ATARI-2600 / Fujiversal-Atari2600-Rev1**. When asking about parts, always give the board's name and revision, which are printed on the board.

Board.	Fujiversal-Atari2600 Rev1
Game chip.	RP2354A (RP2350 with 2 MB flash)
Network chip.	ESP32-S3-WROOM-1, 16 MB flash, 8 MB PSRAM
WiFi.	802.11 b/g/n, 2.4 GHz
Memory card.	microSD, push-push, FAT32
USB.	USB-C: power, and firmware loading
Lights.	status (white / orange); power (red)
Buttons.	RESET; three service buttons (BOOTSEL, S3 EN, S3 BOOT)
Power.	from the console's cartridge port, or USB-C
Games.	up to 32K; .BIN or .ROM; NTSC
Cartridge shell.	3D-printed, after norm8332's "Easy Print" shell
Shell screws.	4 × countersunk wood screw, #4 × 1/2"

Related Publications

- *FujiNet Video Computer System Programmer's Handbook* – writing programs for the cartridge.
- *FujiNet for the Atari 2600: Theory of Operation* – how the cartridge works.
- *M.U.L.E. Player's Guide*, Atari 2600 edition – the whole of M.U.L.E.
- *The FujiNet Network Protocol Handbook* – everything FujiNet can talk to.

A Few Words

WORD	MEANING
Host	A place games are kept: the memory card, or a server on the Internet or in your home.
TNFS	The simple file-sharing language FujiNet servers speak.
Lobby	FujiNet's list of open game tables, on every kind of machine.
Relay	A server that passes two players' moves to each other.
SSID	Your WiFi network's name.
Firmware	The programs built into the cartridge itself.
Page-switching	How a game bigger than 4K shows the console one part of itself at a time. FujiNet does it for the games it loads.

MY FUJINET

Write down your settings here. If you ever start over, everything you need is on this page.

WiFi network (SSID) _____

Password _____

FujiNet's IP address _____

Firmware version _____

Player name _____

HOSTS

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

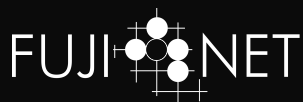
7 _____

8 _____

GAMES I LIKE, AND WHERE THEY ARE

QUICK REFERENCE

SCREEN	JOYSTICK	FIRE	GAME SELECT	GAME RESET
THE FUJINET MENU				
PICK NET	up/down: network	choose it	skip WiFi	—
Keyboard	move the ^	type	ACCEPT · CANCEL · CLEAR	—
FN HOSTS	up/down: host	open it	INFO · RENAME · WIFI · LOBBY	—
FN BROWSE	up/down; right: next page; left: back page / up a folder	open folder · load game	UP DIR · FILTER · INFO · COPY · COPY HERE	—
Any menu			closes the menu	
THE LOBBY				
LOBBY	up/down: table; left/right: page	join	your name	refresh
ENTER NAME	move	type	erase	done
THE GAMES				
5 Card Stud, Texas Hold'em	up/down: move; right: poll; left: menu	make the move	hold: purses	TABLE MENU
Battleship: placing	move the ship	place it	turn it	GAME MENU
Battleship: battle	aim	fire	poll	GAME MENU
Fujitzee	dice; up to the card; up/down rows	roll · hold · score · ready	MENU	poll
M.U.L.E.	walk · choose · bid · declare	start · join · claim · install ...	end turn	hold 1 s: leave
Name screens (games)	move	type; OK finishes	—	Fujitzee: done
SWITCHES				
Left difficulty	At A : Fujitzee shows the standings; M.U.L.E. turns the sound off. In the networked classics each player's own switch counts.			
TV Type	Colour or black and white, as always. In the networked classics it stays your own.			
Cartridge RESET	Back to the FujiNet menu, from anywhere.			



A Worldwide Community Project

FUJINET PROJECT · fujinet.online · github.com/FujiNetWiFi · discord.gg/7MfFTvD